CSCI 2110
DATA STRUCTURES AND ALGORITHMS

TOPICS AND PRACTICE QUESTION BANK FOR FINAL EXAM

NOTE:
FINAL EXAM IS ON SATURDAY, APRIL 11, 2026, 12 NOON
DURATION: 2 HOURS 30 MINUTES
LOCATION: ROWE MANAGEMENT BUILDING POTTER AUDITORIUM (ROOM 1028)
FORMAT: CLOSED BOOK, PROCTORED EXAM ON PAPER

DETAILED BREAKDOWN OF TOPICS

Module 6: BINARY TREES

- Definition and Terminology
- Strictly binary and complete binary trees
- Traversals: Preorder, Inorder, Postorder, Level Order
- Recursive definitions for a binary tree
- Binary Tree Class: Methods and Complexities
- Application: Huffman coding

Module 7: BINARY SEARCH TREES

- Definition and Examples
- Operations – Search, Insert and Delete
- Binary Search Tree Class: Methods and Complexities

Module 8: HEAPS

- Definition and Examples
- Operations – Insert and DeleteMax
- Heap Class: Methods and Complexities
- HeapSort
- Updatable Heaps

Module 9: HASHING AND HASH TABLES

- Motivation
- Hashing, Hash Function and Hash Table Concept with Examples
- Hash Clash
- Solution to Hash Clash: Open Hashing or Separate Chaining
- Solution to Hash Clash: Closed Hashing – Linear Probing and Quadratic Probing
- Miscellaneous Concepts in Hashing
- HashMap in Java with examples

Module 10: GRAPHS

- Definition and Motivation
- Graph Terminology with Examples
- Graph Representation: Adjacency Matrix and Adjacency List
- Graph Algorithms
- Graph Traversals: Depth First and Breadth First
- Topological Sorting

NOTES:

1. For the BinaryTree class, BinarySearchTree class, Heap class, and the HashMap class, you should know how the methods work. For programming questions pertaining to these classes, you will be given the list of methods from the class and their meanings. You may be asked to develop other methods using these methods. You may be given code snippets and asked to fill in the missing lines of code. You may also be given code snippets and asked to trace the output.

3. No cheat sheets or any reference material allowed!

4. You may bring a basic non-programmable calculator – although not needed!

Question 1. Multiple Choice Set - For each of the following questions, select the most appropriate answer:

1. The number of branches from the root to a node in a binary tree is called
    a. height of the node
    b. depth of the node
    c. height of the root
    d. depth of the root
2. A binary tree, in which all levels are filled, except possibly the last level, which is filled from left to right, is called a
    a. Binary Search Tree
    b. Complete Binary Tree
    c. Strictly Binary Tree
    d. Leveled Tree

3. Which of the following statements about binary trees is NOT true?
    a. Every binary tree has at least one node.
    b. Every non-empty tree has exactly one root node.
    c. Every node has at most two children.
    d. Every non-root node has exactly one parent.

The next three questions pertain to a binary tree with 5 nodes. The root is A. A has a left child B and right child C. B has a left child D and a right child E. There are no other nodes.

4. Which of the following traversals yields ABCDE?
    a. Inorder
    b. Level order
    c. Post order
    d. Pre order
    e. Two of the above
5. Which of the following is an inorder traversal of the tree?
    a. ABCDE
    b. ABDEC
    c. DBEAC
    d. DEBAC
    e. None of the above
6. The height of the tree is
    a. 0
    b. 1
    c. 2
    d. 3
    e. None of the above

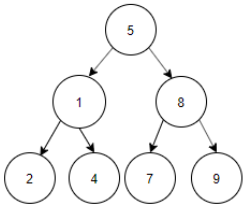
7. Which of the following properties are obeyed by all three tree – traversals?

a. Left subtrees are visited before right subtrees
b. Right subtrees are visited before left subtrees
c. Root node is visited before left subtree
d. Root node is visited before right subtree

8. Suppose you have a binary tree with exactly 7 nodes. What is the maximum height that the tree could have?

a. 2
b. 3
c. 6
d. 7

9. Consider the following binary tree:



Which of the following statements is true?

I. It is a complete binary tree.
II. It is a strictly binary tree.
III. It is a binary search tree.

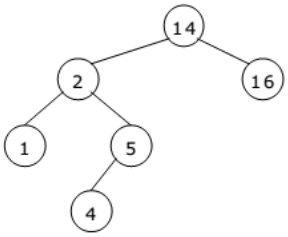
a. I only
b. I and II
c. I, II and III
d. I and III

10. Suppose node with key 7 is NOT THERE in the binary tree in the above question, then which of the following statements is true?

I. It is a complete binary tree.
II. It is a strictly binary tree.
III. It is a binary search tree.

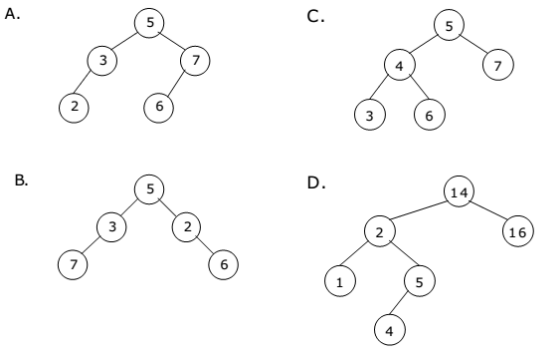
a. I only
b. I only
c. III only
d. I and III
e. None of the statements is true.

11. For the binary search tree shown below, suppose the root is deleted and replaced with **a node in** the left subtree of the root, what will be the new root?



a. 1
b. 2
c. 4
d. 5

12. Which among the following is not a binary search tree?



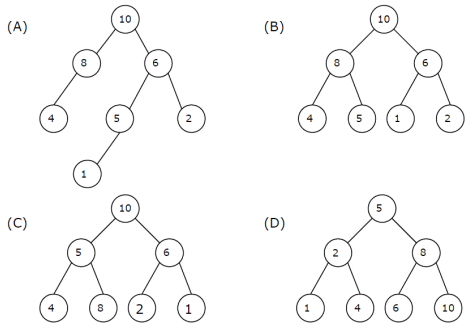
a. A and D
b. B only
c. C only
d. B and C

13. Which of the following statements is true for a binary search tree?

I. It must be a complete binary tree.
II. It must be a strictly binary tree.
III. The key x at a node must be larger than all the keys in the left subtree of the node.
IV. The key x at a node must be smaller than all the keys in the right subtree of the node.

a. I, II and III
b. I, II and IV
c. III only
d. III and IV

14. Which of the following is a binary search tree?



a. A
b. B
c. C
d. D
e. C and D

15. Which of the following statements is true for a heap?

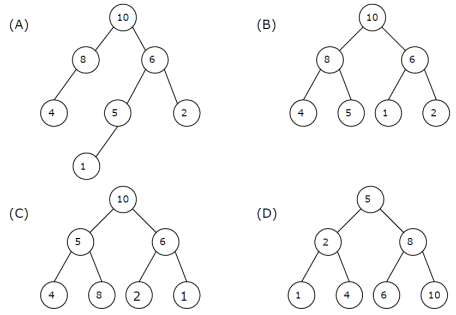
I. It must be a complete binary tree.
II. It must be a strictly binary tree.
III. The key x at a node must be larger than or equal to all the keys in the left and right subtrees of the node.
IV. The key x at a node must be larger than all the keys in the left and right subtrees of the node.

a. I, II and III
b. I, II and IV
c. I and III
d. I and IV

16. What is the worst-case time complexity of building a heap with n items?

a. O(n)
b. O(log n)
c. O(1)
d. O(n log n)

17. Which of the following is a heap?



a. B only
b. B, C and D
c. A and B
d. B and C
e. A and C

18. Consider the following heap:

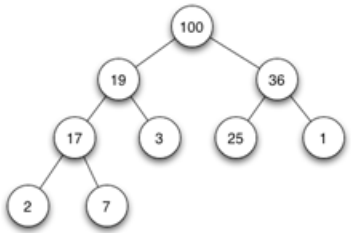


Figure 1

Suppose key = 25 is inserted according to the Sift-up algorithm. What are the keys in the left and right children, respectively, of the root node after the insertion?

a. 19 and 25
b. 25 and 19
c. 25 and 36
d. 36 and 25

19. Consider the same heap shown in Figure 1. Suppose we do a delete_max operation according to the Sift-down algorithm. What are the left and right children of the root node, respectively, after the delete_max?

a. 19 and 25
b. 25 and 19
c. 25 and 36
d. 36 and 25

20. Assuming an arraylist implementation of a heap, which of the following enumerations of an arraylist represents a valid heap? (For keys which are letters/strings, assume that lower the alphabet in lexicographic ordering, lower the priority).

a. [A, B, C, D, E]
b. [10, 10, 10, 10, 10]
c. [20, 10, 5, 9, 21]
d. None of the above

21. Suppose a heap is represented by the following arraylist: [25, 14, 16, 13, 10, 8,12]. What will be the content of the arraylist after two successive delete_max operations using the Sift-down algorithm?
a. [16, 13, 10, 8, 12]
b. [14, 13, 12, 8, 10]
c. [14, 13, 8, 10, 12]
d. [16, 14, 13, 10, 8]

22. Given the following input (4322, 1334, 1471, 9679, 1989, 6171, 6173, 4199) and the hash function x mod 10, which of the following statements are true?

i. 9679, 1989, 4199 hash to the same value
ii. 1471, 6171 hash to the same value
iii. All elements hash to the same value
iv. Each element hashes to a different value

a. i only
b. ii only
c. iii only
d. iv only
e. i and ii

23. Given a hash table T with 25 slots that stores 2000 elements, the load factor α for T is _____

a. 80
b. 0.0125
c. 50,000
d. 1.25

24. A hash table of length 10 uses open addressing with hash function h(k)=k mod 10, and linear probing. After inserting 6 values into an empty hash table, the table is as shown below.

0	
1	
2	42
3	23
4	34
5	52
6	46
7	33
8	
9	

Which one of the following choices gives a possible order in which the key values could have been inserted in the table?

a. 46, 42, 34, 52, 23, 33
b. 34, 42, 23, 52, 33, 46
c. 46, 34, 42, 23, 52, 33
d. 42, 46, 33, 23, 34, 52

25. A hash function h defined h(key)=key mod 10, with linear probing, is used to insert the keys 44, 45, 79, 55, 96, 15 into a table indexed from 0 to 9. What will be the location of key 15?

a. 5
b. 6
c. 7
d. 8

26. In cryptographic hashing, which of the following is correct?

a. The hash of the password is transmitted to the server and the server compares the hash with the stored value.

b. Hashing is not used for password checking.

c. The server sends the hash of the password to the user and the user compares the hash with the stored value.

d. The password is transmitted to the server and the server computes the hash of the password.

27. An undirected graph G has four vertices. Its adjacency matrix representation has all 0's in its **MAIN** diagonal elements and all 1's in its non-diagonal elements. How many edges does the graph G have?

a. 3
b. 6
c. 8
d. 9

28. Which of the following would be the Depth First Traversal of the following graph?

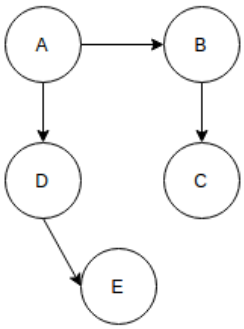


Figure 2

I. ABCDE
II. ADEBC
III. ABDCE
IV. ADBEC

a. I only
b. I and II
c. III only
d. III and IV

29. Consider the graph in Figure 2. How many zeros will be there in the adjacency matrix representation of the graph?

a. 4
b. 5
c. 21
d. 12

30. Consider the graph in Figure 2. The indegree and outdegree of vertex A are, respectively,

a. 1 and 1
b. 2 and 0
c. 0 and 2
d. 1 and 0

“WHO IS LYING?” TYPE MCQs (similar to the ones from attendance quizzes)

MCQ 1
Aanya: “In a strictly binary tree, every node has either 0 or 2 children.”
Ben: “Inorder traversal of any binary tree always gives nodes in sorted order.”
Chloe: “Level order traversal visits nodes level by level from top to bottom.”
Who is lying?
a. Aanya
b. Ben
c. Chloe
d. No one.

MCQ 2
Dev: “In a binary search tree, the left subtree of a node contains values less than the node.”
Esha: “Searching in a balanced BST has O(log n) time complexity.”
Farhan: “Deleting a node in a BST is always an O(1) operation.”
Who is lying?
a. Dev
b. Esha
c. Farhan
d. No one.

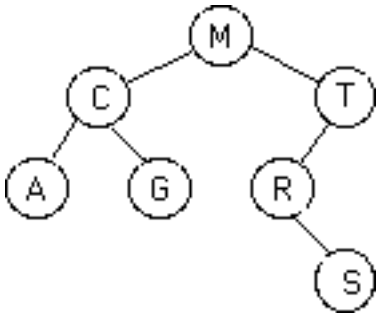
MCQ 3
Gina: “A max heap ensures that the root node contains the maximum value.”
Hassan: “Inserting into a heap may require sifting up to maintain heap property.”
Ishaan: “Deleting the max element from a heap always takes O(1) time.”
Who is lying?
a. Gina
b. Hassan
c. Ishaan
d. No one.

MCQ 4
Jia: “A hash function maps keys to indices in a hash table.”
Karan: “Open hashing handles collisions using linked lists.”
Lina: “Linear probing is one method of closed hashing.”
Who is lying?
a. Jia
b. Karan
c. Lina
d. No one.

MCQ 5
Maya: “Breadth First Search explores nodes level by level.”
Noah: “Depth First Search typically uses recursion.”
Omar: “Topological sorting can be applied to any directed graph.”
Who is lying?
a. Maya
b. Noah
c. Omar
d. No one.

QUESTION NO. 2

You are given the following binary tree:



i) Which are the leaf nodes?

ii) Which are the internal nodes?

iii) What is the longest distance (between any two nodes) in the tree?

iv) Which are the nodes at Level 2?

v) What is the height of the tree?

vi) What are the leaf nodes in the right subtree of the root node?

vii) What is the left subtree of the left child of the right child of the root node?

viii) Is it a strictly binary tree? Why or why not?

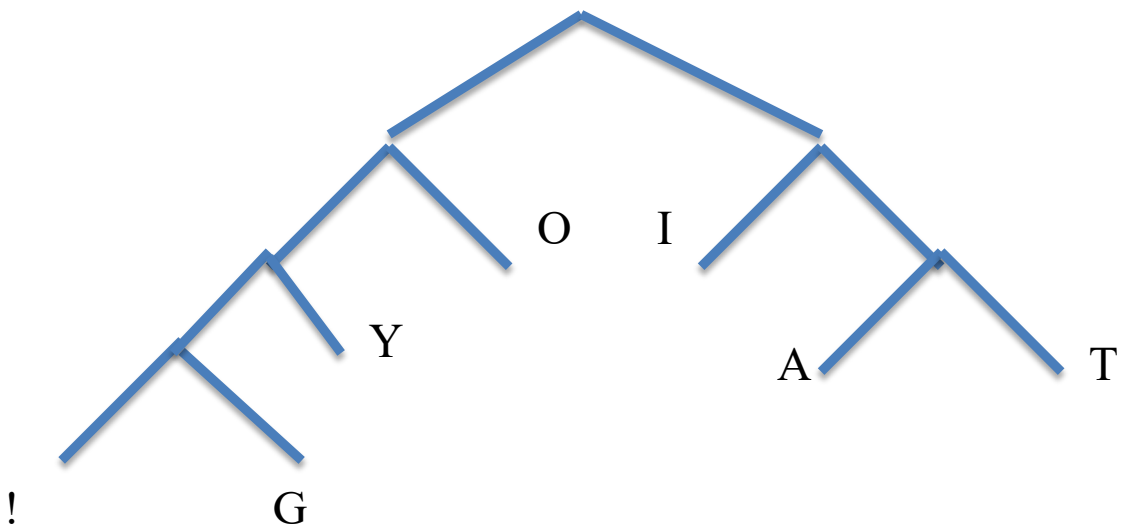
ix) Is it a complete binary tree? Why or why not?

QUESTION NO. 3

3.1 What is the maximum number of nodes in a complete binary tree of height = 3? In general, what is the maximum number of nodes in a complete binary tree of height h?

3.2 What is the minimum number of leaves in a complete binary tree of height = 3? In general, what is the minimum number of nodes in a complete binary tree of height h?

3.3 The following tree was generated using the Huffman technique.



The following message was generated using the codes from the above tree.

0011100010000 10 000101111 1011100000000

Decode the message. Leave the spaces as they are.

QUESTION NO. 4

For the following programming questions, refer to the Binary Tree class methods given in the reference sheet at the end.

4.1 Write a class called BinaryTreeUtils that has the following three methods:

a) A recursive method to count the number of nodes in a binary tree.

b) A recursive method that compares two binary trees and returns true if they are identical. Two binary trees are identical if they have the same structure and the same data in each corresponding node.

```
public class BinaryTreeUtils {

  public static <T> int countNodes(BinaryTree<T> tree) {









  }

  public static <T> boolean areIdentical(BinaryTree<T> tree1, BinaryTree<T> tree2) {









}
}
```

4.2 Assume that the following code snippet is run to construct a binary tree.

```
BinaryTree<String> root = new BinaryTree<>();
root.makeRoot("A");

BinaryTree<String> B = new BinaryTree<>();
B.makeRoot("B");
BinaryTree<String> C = new BinaryTree<>();
C.makeRoot("C");

BinaryTree<String> D = new BinaryTree<>();
D.makeRoot("D");

root.attachLeft(B);
root.attachRight(C);
B.attachLeft(D);
```

Draw the binary tree that is created with the above code.

Now assume that the following code snippet is executed.

```
BinaryTree<String> X = root.detachLeft();
BinaryTree<String> E = new BinaryTree<>();
E.makeRoot("E");
X.attachRight(E);
root.attachLeft(x);
```

Draw the tree structure after the above code snippet is executed.

QUESTION NO. 5

Starting from an empty **binary search tree,** suppose that the following operations are performed ***in sequence***.

- Insert 55, 23, 48, 95, 16, 2, 70, 63, 100, 19

a) What is the root node?

b) What are the leaf nodes?

c) What are the left and right child nodes of the root?

- Delete the right child of the root node

a) What are the leaf nodes?

b) What are the left and right child nodes of the root?

- Delete the left child of the root node

a) What are the leaf nodes?

b) What are the left and right child nodes of the root?

- Delete the root node

a) What is the new root node?

b) What are the leaf nodes?

c) What are the left and right child nodes of the root?

QUESTION NO. 6

Starting from an empty heap, do the following operations *in sequence.*

a) Insert 12, 19, 10, 4, 23, 7, 45, 8, 15

b) Delete_max

c) Insert 22
d) Delete_max
e) Delete_max five times.

Assuming an arraylist implementation of the heap, write the enumeration of the arraylist after each operation.

QUESTION NO. 7

7.1 The keys 12, 18, 13, 2, 3, 23, 5 and 15 are inserted into an initially empty hash table of length 10 using closed hashing and linear probing. The hash function is h(k) = k mod 10.

What is the resultant hash table? Write your answer as a sequence of numbers separated by commas and a '-' for an empty array slot. Alternatively, you can draw the table.

For example, if your resulting hash table is the array (Note: This is not the correct answer!)

12		13		18	2	3	23	5	15

You would write it your answer as follows
12, - , 13, - , 18, 2, 3, 23, 5, 15

7.2 Repeat the above question but this time use quadratic probing.

7.3 Consider a hash table of size seven, with starting index zero, and a hash function (3x + 4) mod 7, where x is the key. Assuming the hash table is initially empty, the sequence of keys 1, 3, 8, 10 is inserted into the table using closed hashing and linear probing?

What is the resultant hash table? Write your answer as a sequence of numbers separated by commas and a '-' for an empty array slot.

7.4 The figure below shows the current state of a 10-element hash table, with the occupied cells shown with an XX. Assuming that the hash function is key mod 10. The following keys are inserted in sequence:

44, 55, 66, 77

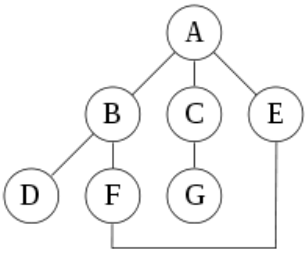
	XX		XX	XX	XX	XX	XX		
0	1	2	3	4	5	6	7	8	9

a) What is the resulting table if Linear probing is used? What is the total number of collisions?

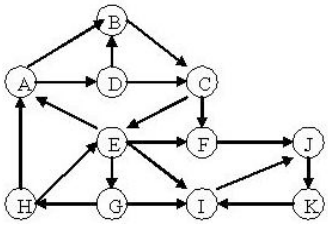
b) What is the resulting table if Quadratic probing is used? What is the total number of collisions?

QUESTION NO. 8:

For the graph given below
a) State whether the graph is weighted or unweighted, directed or undirected, connected or unconnected, cyclic or non-cyclic.
b) List all vertices with degree = 3.
c) List a cycle in the graph, if it exists.
d) List all possible simple paths from A to D.
e) Determine the DFS and BFS traversals from node A.

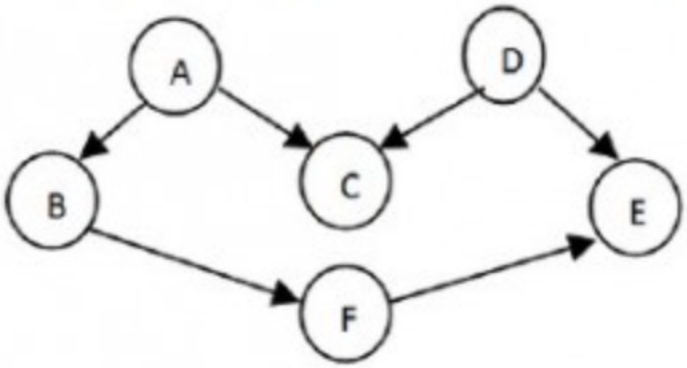


QUESTION NO. 9:
For the following graph, which node(s) has (have) the highest indegree? What is the highest indegree? Which node(s) has (have) the highest outdegree? What is the outdegree? Is the subgraph connecting vertices A, B, C, D strongly connected or weakly connected?



QUESTION NO. 10

For the graph shown below, find one solution that lists the vertices so that the dependency is not violated. (Run the topological sorting algorithm and assign topological numbers to the vertices. Show steps).



Binary Tree Class

public class BinaryTree<T>

Instance Variables

T data;
BinaryTree<T> parent;
BinaryTree<T> left;
BinaryTree<T> right;

Constructor

BinaryTree<T> ()	Creates an empty binary tree

Methods

Name	What it does	Header
makeRoot	Creates a root node for this tree with the specified data	void makeRoot(T data)
set methods	sets the data, left, right and parent	void setData(T data) void setLeft(BinaryTree<T> tree) void setRight(BinaryTree<T> tree) void setParent(BinaryTree<T> tree)
get methods	Gets the data, left, right and parent	T getData() BinaryTree<T> getLeft() BinaryTree<T> getRIght() BinaryTree<T> getParent()
root	Returns the root of the tree in which this tree is a subtree	BinaryTree<T> root()
attachLeft	Attaches the specified tree as the left subtree of this tree node.	void attachLeft(BinaryTree<T> tree)
attachRight	Attaches the specified tree as the right subtree of this tree node	void attachRight (BinaryTree<T> tree)
detachLeft	Detaches the left subtree from this tree node and returns it	BinaryTree<T> detachLeft()
detachRight	Detaches the right subtree from this tree node and returns it	BinaryTree<T> detachRight()
isEmpty	Returns true if this tree is empty, false otherwise	boolean isEmpty()
clear	Empties out this binary tree and detaches it from its parent node	void clear()

BinarySearchTree Class

public class BinarySearchTree<<T extends Comparable<T>>

Instance Variables

BinaryTree<T> tree;
int size;

Constructor

BinarySearchTree()	Creates an empty binary tree

Methods

Name	What it does	Header
getTree	Returns the tree	BinaryTree<T> getTree()
isEmpty	Checks if the tree is empty	boolean isEmpty()
Size	Returns the size of the tree	Int size()
search	Searches the tree for a given key and returns the node with the key	BinaryTree<T> search(T key)
Insert	Inserts a given key into the binary search tree	void insert(T key)
Delete	Deletes a node with the given key	void delete(T key)